

offline

❶ CCpR Measurement

Single-point load injection Eq.(4):

$$\mathrm{CCpR}_{I\times S} = \begin{bmatrix} 3.1 & \cdots & 9.6 \\ \vdots & \ddots & \vdots \\ 0.5 & \cdots & 7.3 \end{bmatrix}$$

from Service S_1 to S_s; from API /A to /I

online

❷ Metrics Preprocessing

Merge redirect chain Eq.(6)(7):

$$/A_1 \xrightarrow{302} /A_2 \xrightarrow{302} \cdots \xrightarrow{302} /A_n \xrightarrow{200}$$

$$/A_1 \xrightarrow{200}$$

❸ Workloads Quantification

Closed-loop Little's Law Eq.(3):

$$\mathrm{Workload}_{1\times I} = \Sigma \mathrm{RPS}\cdot(\mathrm{AIL}+\mathrm{AOL}) = \underbrace{\begin{bmatrix} 30 & \cdots & 50 \end{bmatrix}}_{\text{from API /A to /I}}$$

❹ Target Throughput Calculation

Throttling equation Eq.(8):

$$\mathrm{RPS}^*_{1\times I} = \frac{\mathrm{Workload}_{1\times I}}{(\mathrm{AIL}+\mathrm{SLO})} = \underbrace{\begin{bmatrix} 26 & \cdots & 43 \end{bmatrix}}_{\text{from API /A to /I}}$$

❺ Instance Count Calculation

Eq.(9): $\mathrm{Allocation}_{1\times S}$

$$= \left\lceil \frac{\mathrm{RPS}^*_{1\times I} \times \mathrm{CCpR}_{I\times S}}{\text{CPU request}} \right\rceil = \underbrace{\begin{bmatrix} 7 & \cdots & 12 \end{bmatrix}}_{\text{from Service } S_1 \text{ to } S_s}$$